Venkata Sai

Software Engineer

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Professional Experience

Software Engineer *Jul 2025 – Present*

Persistent Systems | AI-Powered Customer Support Systems

- Designed and implemented a **GenAI Bot** for customer support, leveraging **LangChain** and **OpenAI API** on **AWS Lambda** to improve response times by 30% for complex queries.
- Led development of scalable **Python** microservices on **Kubernetes** for AI backend, processing 5M daily user interactions while maintaining 99.9% system reliability.
- Optimized **SQL** database schemas and queries for the **GenAI Bot** knowledge base, reducing data retrieval latency by 40% and enhancing user experience significantly.
- Implemented advanced **monitoring** and **debugging** tools with **Prometheus** and **Grafana** across **distributed systems**, proactively identifying and resolving production issues within 15 minutes.
- Architected a resilient data pipeline using **Kafka** and **Spark** for real-time analytics on customer support data, improving decision-making speed by 25% for management.
- Mentored junior engineers on **Software Development Lifecycle** (SDLC) best practices and **Object Oriented Programming** (OOP) principles, boosting team's code quality by 20% consistently.
- Contributed to designing new **technology stacks** for future product features, integrating **AWS S3** for data storage, reducing infrastructure costs by 15% annually.

Software Engineer *Jun 2024 – Jun 2025*

Nagarro | Enterprise Data Integration Platform

- Developed high-performance **Python** ETL jobs on **Apache Airflow** for ingesting diverse enterprise data, processing 10M records hourly with 99.8% data accuracy.
- Built and maintained **Java** microservices using **Spring Boot** within a **distributed systems** architecture, improving application responsiveness for 50,000 active users.
- Implemented robust **data structures and algorithms** in **C++** for real-time analytics engines, accelerating complex data computations by 35% across various modules.
- Designed and optimized **MySQL** databases for critical business applications, enhancing query performance by 25% and supporting daily operations for 20+ teams.
- Utilized **GIT** for version control and collaborative development, enforcing rigorous **Software Development Lifecycle** (SDLC) processes, reducing merge conflicts by 30%.
- Implemented comprehensive unit and integration tests, ensuring high code quality and system reliability across **software components**, decreasing production defects by 18%.
- Monitored and debugged production services using **Splunk** and **Datadog**, resolving critical issues within an average of 30 minutes to minimize downtime.
- Developed custom APIs using **Python Flask** for internal tools, integrating various **software applications** and boosting engineering productivity by 20% for developers.

Associate Software Engineer *Jul 2021 – Aug 2022*

Virtusa | Web Application Development

- Developed core modules for client-facing **Java** web applications, applying **Object Oriented Programming** (OOP) principles and improving feature delivery speed by 15%.
- Implemented efficient **data structures and algorithms** in **C++** for backend services, optimizing data processing and query response times by 20%.
- Designed and managed **SQL** database schemas for various projects, ensuring data integrity and supporting robust **software applications** for over 10,000 users.
- Participated in full **Software Development Lifecycle** (SDLC) activities, from requirements gathering to deployment, enhancing team collaboration and project success rate by 10%.
- Utilized **GIT** for source code management and collaborated effectively within agile teams, ensuring timely delivery of critical features to meet project deadlines 95% of the time.

Skills

Languages: Python, Java, C++
Frameworks & Tools: LangChain, Kubernetes, Prometheus, Grafana, Kafka, Spark, Apache Airflow, Spring Boot, GIT, Splunk, Datadog, Python Flask, GenAI Bot, OpenAI API
Cloud & Infra: AWS Lambda, AWS S3, SQL, MySQL, Distributed Systems
Concepts: Software Development Lifecycle (SDLC), Object Oriented Programming (OOP), Data Structures & Algorithms, Debugging, Monitoring, Technology Stacks, Software Components, Software Applications

Education

Master's in Computer Science, Oklahoma State University, Stillwater
Bachelor's in Computer Science, JNTU Hyderabad, India